

Source: Calculated using data from Robert Shiller's website.

With the correlated error terms, inflation is modeled as a first-order autoregressive process starting from 1.7 percent inflation at the end of 2016 and trending toward its historical average of 2.9 percent over time with its historical volatility. Bond yields are similarly modeled with a first-order autoregression with an initial seed value of 2.45 percent, representing the yield for ten-year Treasury bonds at the start of 2017. Next, home prices and the risk premium are both modeled as random walks around their historical averages and with their historical volatilities. Bond returns are calculated from bond yields and changes in interest rates, assuming a bond mutual fund with equal holdings of past ten-year Treasury issues. Stock returns are calculated as the sum of bond yields and the simulated equity risk premium over yields. Exhibit 3A.2 shows the simulated median outcomes for the key variables using one hundred thousand simulations. As can be seen, stock and bond returns start lower but trend upward, on average, toward their historical averages.

Exhibit 3A.2

Medians of Simulated Outcomes for Inflation, Bonds, and Stocks

